Laboratory Guide 2: Entanglement & Multi-Qubit Systems

The "Spooky Action at a Distance"

Course: Quantum Computing Foundation

1 Introduction and Objectives

In Lab 1, we manipulated individual qubits. In this laboratory, we explore systems of multiple qubits and the phenomenon of **Entanglement**. As stated in Lesson QC05, the real power of quantum computation appears when we start combining qubits together.

Prerequisites:

- Lesson QC05: *Multiple Qubits*
- Lab 1: *Introduction to QISKIT*

Learning Objectives:

1. Understand the Controlled-NOT (CNOT) gate logic.
2. Create a Bell State (Maximally Entangled State) using the Composer.
3. Visualize correlations between qubits on the Q-Sphere.
4. Verify entanglement statistics using Qiskit code.
5. Extend entanglement to 3 qubits (GHZ State) and discuss scalability.

Important: Qiskit Bit Ordering

Reading Order in Qiskit
Qiskit uses "Little Endian" ordering. This means the rightmost bit is qubit 0 (q_0).

$$\text{State } |10\rangle \implies q_1 = 1, \quad q_0 = 0$$

Always pay attention to which qubit is the *Control* and which is the *Target*.

2 Part A: Logic & Entanglement (IBM Quantum Composer)

Platform: IBM Quantum Composer

2.1 Exercise 1: The CNOT Gate (Conditional Logic)

Context: Lesson **QC05** introduces the CNOT gate. It acts on two qubits: a *Control* and a *Target*. It flips the Target qubit ONLY IF the Control qubit is in state $|1\rangle$.

Student Activity

Action:

1. Use an `X` gate on q_0 (Control) to set it to $|1\rangle$. Leave q_1 (Target) at $|0\rangle$.
2. Drag a `CNOT` gate (symbol $\oplus$ connected to a dot). Make sure the **dot (control)** is on q_0 and the **plus (target)** is on q_1.
3. Observe the Output State.

Expected Effect: The state becomes $|11\rangle$ (Probability 100% on "11"). The target flipped because the control was 1.

2.2 Exercise 2: Creating the Bell State (Entanglement)

Context: Entanglement is a state where two particles become inextricably linked. The most famous example is the **Bell State** $|\Phi^+\rangle = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$.

Student Activity

Action:

1. Clear the circuit.
2. Apply an `H` gate to q_0. (State is now superposition of 0 and 1).
3. Apply a `CNOT` gate (Control q_0, Target q_1).
4. Check the **Probabilities**.

Expected Effect: You should see two bars:

- $\approx 50\%$ Probability for $|00\rangle$
- $\approx 50\%$ Probability for $|11\rangle$
- **0% Probability** for $|01\rangle$ or $|10\rangle$.

IBM Composer View (Q-Sphere)

Q-Sphere Visualization
Look at the Q-Sphere. You will see two dots connected or symmetric. Unlike two separate qubits in superposition (which would show 4 dots: 00, 01, 10, 11), an entangled state only shows the correlated outcomes (00 and 11).

Deep Dive: Why does this matter?

Why is Entanglement so Important?
Entanglement is not just a weird phenomenon; it is the fundamental resource for two game-changing applications.

1. Computational Power (The "Speedup")
As we will see in **Lesson QC14** (Algorithms), entanglement allows a quantum computer to process a vast state space simultaneously. Here a brief description to some of those algorithms, that extensively makes use of the entanglement effect:

- **Shor's Algorithm:** Uses entanglement to crack RSA encryption exponentially faster than classical computers.
- **Grover's Algorithm:** Uses entanglement to search through unsorted databases with quadratic speedup.
- *Hardware Context:* Chips like **Google Willow** (2025) use 70+ entangled qubits to perform tasks impossible for classical supercomputers.

2. Unhackable Security (Quantum Internet)
Entanglement is the basis of **Quantum Key Distribution (QKD)**.

- If two parties share entangled photons, any attempt by a hacker to "eavesdrop" (measure the line) destroys the entanglement instantly.
- This collapse reveals the intruder immediately, guaranteeing physical security.
- *Hardware Context:* Companies like **IonQ** and **Quantinuum** are building the nodes for these quantum networks.

3 Part B: Coding Entanglement (Qiskit)

3.1 Exercise 3: The "Hello World" of Quantum Code

We will now write the Python code to generate the Bell State. This is the standard test to verify a quantum computer works.

Coding Task: Bell State

```python
from qiskit import QuantumCircuit
from qiskit_aer import AerSimulator
from qiskit.visualization import plot_histogram

# 1. Initialize Circuit with 2 Qubits
qc = QuantumCircuit(2, 2)

# 2. H on q0, then CNOT(0->1)
qc.h(0)
qc.cx(0, 1)

# 3. Measure both
qc.measure([0, 1], [0, 1])

# 4. Execute
backend = AerSimulator()
job = backend.run(qc, shots=1000)
print(f"Bell Results: {job.result().get_counts()}")
```

3.2 Exercise 4: Scaling Up - The GHZ State (3 Qubits)

Context: State-of-the-art quantum processors (e.g., **Google Willow**, **IonQ Tempo**) rely on creating entanglement across many qubits. The simplest example of *multipartite entanglement* is the **GHZ State**:

$$|GHZ\rangle = \frac{|000\rangle + |111\rangle}{\sqrt{2}}$$

Concept Preview: What is Decoherence?

You will study this formally in **Lesson QC06**, but for now, think of **Decoherence** as "environmental noise".

- Just as a spinning top eventually falls due to friction, quantum states lose their superposition due to interaction with the outside world (heat, radiation).
- **The Rule:** The more qubits you entangle, the faster the system decoheres. This is why a 3-qubit GHZ state is much more fragile than a 2-qubit Bell state.

Coding Task: The 3-Qubit Entangler

```python
# 1. Initialize 3 qubits
qc3 = QuantumCircuit(3, 3)

# 2. Entangle q0 and q1 (Bell State)
qc3.h(0)
qc3.cx(0, 1)

# 3. Entangle q2 (Chain reaction: q1 controls q2)
qc3.cx(1, 2)

# 4. Measure
qc3.measure([0, 1, 2], [0, 1, 2])

# 5. Execute
job3 = AerSimulator().run(qc3, shots=1000)
print(f"GHZ Results: {job3.result().get_counts()}")
```

Expected Analysis: Ideal results are '000' and '111'. In a real system, you would see errors (e.g., '001') appearing more frequently here than in the previous exercise, exactly due to the decoherence effect described above.

4 Part C: Assessment (Quiz)

Q1. The Logic of CNOT

If the control qubit is in state $|0\rangle$ and the target qubit is in state $|1\rangle$, what does the CNOT gate do?

A) Flips the target to $|0\rangle$.

B) Flips the control to $|1\rangle$.

C) Nothing. The state remains $|01\rangle$ (Control is 0, so Target is untouched).

Q2. Instantaneous Correlation

You have created the Bell State $\frac{|00\rangle+|11\rangle}{\sqrt{2}}$. You measure the first qubit and get the result **"1"**. What is the state of the second qubit immediately after?

A) Still 50% "0" and 50% "1".

B) Definitely "1". The wavefunction has collapsed to $|11\rangle$.

C) Definitely "0".

Q3. The Fragility of Large Systems

In Exercise 4, we mentioned that "Decoherence" acts like noise. Why is the 3-qubit GHZ state harder to maintain than a single qubit?

A) Because Qiskit cannot handle 3 qubits.

B) Because having more entangled particles increases the interaction surface with the environment, making the system lose information faster.

C) Because 3 is an odd number.

Solutions: Q1: C; Q2: B; Q3: B (More qubits = Faster decoherence).